The Greenhouse Effect

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The greenhouse effect

Temperature of Earth

- If the Earth had no atmosphere, the average temperature on the surface would drop to about –18 °C
  - This would happen because the surface would be emitting **all** the radiation from the Sun into space
- The temperature of the Earth is affected by factors controlling the balance between **incoming** radiation and radiation **emitted**
- The Earth receives the majority of its heat in the form of thermal radiation from the Sun
  - At the same time, the Earth **emits** its own thermal radiation, with a slightly longer wavelength than the thermal radiation it receives (this is caused by difference in surface temperature of the Earth and the Sun)

The greenhouse effect

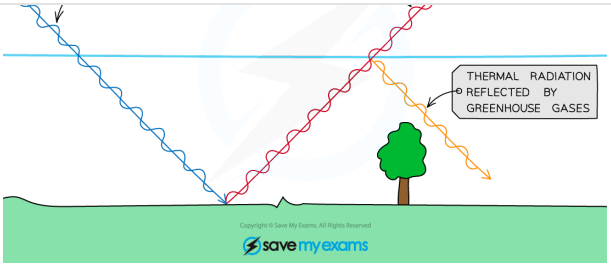
- Some gases in the atmosphere, such as water vapour, methane, and carbon dioxide (greenhouse gases) absorb and reflect back longer-wavelength infrared radiation from the Earth and prevent it from escaping into space
  - These gases **absorb** the radiation and then **emit** it back to the **surface**
  - These gases keep the Earth's temperature at a reasonable level, and the process is called **the greenhouse effect**
- This process makes the Earth warmer than it would be if these gases were not in its atmosphere

Radiation and the greenhouse effect

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



The Earth receives thermal radiation from the Sun but emits its own thermal radiation at the same time. The greenhouse effect is where this radiation is trapped by gases

- The temperature of the Earth, therefore, depends on several factors, such as the rate that infrared radiation is:
  - **Reflected** back into space
  - **Absorbed** by the Earth's atmosphere or by the Earth's surface
  - **Emitted** from the Earth's surface and from the Earth's atmosphere into space

Worked Example

Explain why adding an excess of carbon dioxide to the Earth's atmosphere would lead to an increase in the surface temperature of Earth.

Answer:

Step 1: Describe the greenhouse effect

- Radiation from the Sun is absorbed by the Earth
- Some is re-emitted by the Earth
- Greenhouse gases, such as carbon dioxide, reflect a fraction of this emitted radiation and heat the Earth's surface

Step 2: Describe the effect of increasing the amount of carbon dioxide

- If there is more carbon dioxide, more of the Earth's emitted radiation is reflected back to the Earth
- The rate of radiation absorbed by the Earth's surface increases, so its temperature increases

Examiner Tips and Tricks

Global warming is an important current topic in the world. Generally it is important to understand how this process works, but this also means it appears in exams frequently. Learning more about this topic not only means that you are prepared for application questions on radiation, but it also

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